Volcano Eruption Prediction from Earthquake Data

UE24CS352A Machine Learning Mini-Project (Project No. 64) | Vishwas V Angadi (PES1UG24AM416), Tasya Shetty (PES1UG24AM411)

Code: github.com/PES1UG24AM411/Predicting-_Eruptive_Events_at_Volcanoes_from_Earthquake_Data

1. Problem Statement

Can earthquake activity at Kilauea volcano (Hawaii) tell us whether the volcano is erupting, and how long until the next eruption? We study two tasks on the Pu'u 'O'o earthquake catalog, following the Stanford CS229 PEEVED project: **(1) classification**: was Kilauea erupting when an earthquake occurred, and **(2) regression**: how many days remain until the next eruption (0 during an eruption). The aim is to build and honestly evaluate an ML pipeline, not to claim a deployable forecasting system.

2. Dataset

- **puuoo_earthquakes.csv**: 6,485 earthquakes from 1 Jan 1983 to 30 Dec 1986 (time, latitude, longitude, depth, magnitude). Median depth 7.8 km, median magnitude 1.9.
- **PuuOo.csv** (eruption labels): 49 Pu'u 'O'o episodes with start date and length in hours. The authors' Excel workbook has an irregular layout that the loader cannot parse, so this file is used. One entry (episode 48, the long Kupaianaha eruption) holds 173,566,800 labelled 'hrs'; this is really seconds (about 5.5 years), so it is converted to hours.
- **Labels:** 951 of 6,485 earthquakes (14.66%) occur during an eruption, so the classes are imbalanced. Time-to-eruption (tte) is at most 63.9 days, with a median of 12.9 days outside eruptions.

3. Approach

- **Features (7):** longitude, latitude, depth, magnitude, and earthquake counts in the previous 1, 7 and 30 days (median 5, 28 and 111). Counts use only earlier events, so the current or future earthquakes never leak into the features.
- **Splits:** a random stratified 70/20/10 split (4,539 / 1,297 / 649; paper-style) and a chronological 70/20/10 split as a robustness check, since nearby earthquakes are correlated in time.
- **Models:** classification uses Logistic Regression, K-Means (nearest class centroids, k=8), Random Forest and a Neural Network. Regression uses K-Means (k=10), Random Forest and a Neural Network (4 hidden layers of 256, early stopping). Random Forest trees and depth are tuned on the dev split (classification: 300 trees, depth 20).
- **Imbalance handling:** balanced class weights for Logistic Regression and the Random Forest, and a decision threshold chosen on the dev set only to maximise Cohen's kappa. The regression target is standardised with training data only.
- **Metrics:** AUROC, Cohen's kappa, precision, recall, F1, accuracy and confusion matrix for classification; RMSE (days) and R^2 for regression. Accuracy alone is misleading because 85% of random-test earthquakes are not erupting.

4. Implementation Overview

The project is a Python package. `src/data.py` loads the files, builds the labels and the 7 features and creates both splits; `src/models.py` defines the models, Random Forest tuning, threshold tuning and evaluation; `src/plots.py` saves the plots to `results/`. `main.py` runs the whole pipeline end-to-end (`python main.py`), and the notebook in `notebooks/` walks through the same steps with figures. Dependencies are in `requirements.txt`.

5. Results (test set, real data)

Classification, random split (649 test earthquakes, 95 erupting)

Model	AUROC	Kappa	Precision	Recall	F1	Accuracy
Random Forest	0.894	0.583	0.674	0.611	0.641	0.900
Neural Network	0.851	0.431	0.629	0.411	0.497	0.878
Logistic Regression	0.795	0.288	0.373	0.432	0.400	0.810
K-Means	0.691	0.115	0.204	0.716	0.317	0.549

The Random Forest finds 58 of the 95 erupting earthquakes with 28 false alarms (TN 526, FP 28, FN 37, TP 58). Balanced weights and dev-tuned thresholds raised kappa from 0.00 to 0.29 for Logistic Regression (which previously predicted 'not erupting' every time) and from 0.53 to 0.58 for the Random Forest, with AUROC essentially unchanged.

Regression, random split

Model	RMSE (days)	RMSE (scaled)	R^2
Random Forest	9.65	0.714	0.505
Neural Network	10.10	0.748	0.457
K-Means	13.76	1.019	-0.007

Chronological split (robustness check)

Result	Random split	Chronological split
Share of test earthquakes erupting	14.6%	74.6%
Classification AUROC (all models)	0.69 to 0.89	0.45 to 0.47
Classification kappa (all models)	0.12 to 0.58	-0.07 to -0.02
Regression R^2	-0.01 to 0.51	-2.1 to -3.0

Feature importance (Random Forest): the 30-day earthquake count is the most important feature for both tasks (0.29 for classification, 0.30 for regression), followed by the 7-day count (0.18 and 0.16). Importance shows which inputs the model uses; it does not show causation.

6. Conclusions

- Within one period, earthquake-derived features separate erupting from non-erupting earthquakes well (Random Forest AUROC 0.89, kappa 0.58) and give moderate skill for time-to-eruption (R^2 0.51, about 9.6 days RMSE). The Random Forest is the best model on both tasks.
- On a chronological split the skill disappears (AUROC about 0.5, R^2 below 0). The later period is dominated by one long eruption, unlike the short repeating episodes seen in training, and random splits let neighbouring earthquakes leak between train and test. Earthquake data alone is therefore not enough for reliable forecasting.
- **Limitations:** a single volcano, 1983 to 1986 only, and labels from one eruption chronology. **Next steps:** add deformation and gas measurements, evaluate only on future periods, and report uncertainty.